

Goals:

- Use DB for metadata and fast searching
- Use S3 mainly for uploaded files or attachments and for data backups.

Creating DB,

I chose MySQL.

Master Username: admin

Password: CompanyNetwork123

Aurora and RDS > Databases > Create database

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource

Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ Connect to an EC2 compute resource

Set up a connection to an EC2 compute resource for this database.

EC2 instance

Info

Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

i-Ocb6dfdc31bfb74e0

▼

Private-Server

ⓘ

Some VPC settings can't be changed when a compute resource is added

Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group `rds-ec2-X` is added to the database and another called `ec2-rds-X` to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.

Network type

Info

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ IPv4

Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode

Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC)

Info

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

VPC-Company-Production-Network (vpc-0d45c7156eb19449c)

▼

2 Subnets, 1 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

Aurora and RDS > Databases > Create database

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No

RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall)

Info

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing

Choose existing VPC security groups

☐ Create new

Create new VPC security group

Additional VPC security group

Choose one or more options

▼

Private-Server

×

ⓘ

Amazon RDS will add a new VPC security group `rds-ec2-1` to allow connectivity with your compute resource.

Availability Zone

Info

us-east-1a

▼

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ Create an RDS Proxy

Info

RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

▼ Additional monitoring settings

Enhanced Monitoring, CloudWatch Logs and DevOps Guru

Enhanced Monitoring

☐ Enable Enhanced monitoring

Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Log exports

Select the log types to publish to Amazon CloudWatch Logs

☒ Audit log

☒ Error log

☒ General log

☐ iam-db-auth-error log

☐ Slow query log

IAM role

The following service-linked role is used for publishing logs to CloudWatch Logs.

RDS service-linked role

Initial database name: Give it a name like `company_site`. If you leave this blank, the database starts empty with no "folder" to put tables in

Private instance/machine:

```
sudo apt update
```

```
sudo apt install mysql-client -y
```

Creating subnet group:

You need to create this new subnet group so that you have another private subnet to store your database and S3 bucket for storage.

☰ [Aurora and RDS](#) > [Subnet groups](#) > Create DB subnet group

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name

You won't be able to modify the name after your subnet group has been created.

company-db-group

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

Company network for storing DB and S3 bucket

VPC

Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

VPC-Company-Production-Network (vpc-0d45c7156eb19449c)

7 Subnets, 5 Availability Zones

[Aurora and RDS](#) > [Subnet groups](#) > Create DB subnet group

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone ▼

us-east-1a ✕
 us-east-1b ✕
 us-east-1c ✕
 us-east-1f ✕

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets ▼

RDS-Pvt-subnet-3
Subnet ID: subnet-03edd76655489700d CIDR: 10.0.3.128/25 ✕
 RDS-Pvt-subnet-5
Subnet ID: subnet-0a2754a2c51817f82 CIDR: 10.0.4.128/25 ✕

RDS-Pvt-subnet-1
Subnet ID: subnet-03429d3a6d1e36db8 CIDR: 10.0.0.128/25 ✕

ⓘ For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (3)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1b	RDS-Pvt-subnet-3	subnet-03edd76655489700d	10.0.3.128/25
us-east-1f	RDS-Pvt-subnet-5	subnet-0a2754a2c51817f82	10.0.4.128/25
us-east-1a	RDS-Pvt-subnet-1	subnet-03429d3a6d1e36db8	10.0.0.128/25

Back to creating a database,

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

☒ Choose existing
Choose existing DB subnet group
 ☐ Automatic setup
RDS creates a new subnet group for you or reuse

Existing DB subnet groups

company-db-group
3 Subnets, 3 Availability Zones ▼

Public access [Info](#)

Database - How to connect to your RDS

Since your `mysql-client` is already installed, you are ready to go. You just need to tell the client where your RDS "TV" is.

1. **Get your Endpoint:** Go to the AWS RDS Console, click on your database, and look for the **Endpoint**. It will look something like:
`company-db.xyz.abc.us-east-1.rds.amazonaws.com`

Aurora and RDS > Databases > company-db

Connect using **Info**

☐ Code snippets
Use when connecting through SDK, APIs, or third-party tools including agents.

☐ CloudShell
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☒ Endpoints
Use when connecting through any IDE interface.

Database name
companyDBTest

Master username
admin

Internet access gateway
Disabled

Port
3306

Endpoint type
Instance endpoint

Additional configurations

Connectivity & security

Endpoint & port	Networking	Security
Endpoint company-db.c2lqwc4e8qkg.us-east-1.rds.amazonaws.com	Availability Zone us-east-1a	VPC security groups rds-ec2-2 (sg-01d146b4868dffc4) Active rds-ec2-1 (sg-0ec44dd70fb813d68) Active
Port 3306	VPC VPC-Company-Production-Network (vpc-0d45c7156eb19449c)	Publicly accessible No
	Subnet group	

Run this command from your Private Instance:

```
mysql -h [YOUR_RDS_ENDPOINT] -P 3306 -u admin -p
```

2. (Replace *[YOUR_RDS_ENDPOINT]* with your actual link and *admin* with the username you chose).
3. **Enter your Password:** When prompted, type the password you set during the RDS creation. (Note: The cursor won't move while you type—this is a security feature).

🧱 If it hangs (The Connection Test)

If the command just sits there and never asks for a password, your **Security Group** is blocking the door.

Quick Check: Run this command on your Private Instance to see if the "pipe" is even open:

```
telnet [YOUR_RDS_ENDPOINT] 3306
```

- **If it says "Connected":** The firewall is fine. You just had a typo in your password or username.
- **If it says "Connection timed out":** You need to go to the **RDS Security Group** and add an **Inbound Rule** for Port **3306** from the IP or Security Group of your Private Instance.

This is on a public instance (reverse proxy):

```
ubuntu@ip-10-0-1-252:~$ mysql -h company-db.c2lqwc4e8qkg.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:

```

It is stuck here cuz the database is only set to connect to the private instance.

This is on a private instance.

```
ubuntu@ip-10-0-2-37:~$ mysql -h company-db.c2lqwc4e8qkg.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 35
Server version: 8.4.7 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> 
```

If you want to know where you are while you are inside the database, use this command:

```
SELECT DATABASE();
```

To get out of MySQL:

```
EXIT;
```

DB CLI tips

The moment you see mysql>, type this command to see all folders:

```
SHOW DATABASES;
```

Then, select the first one as it is created when you create the database in the console.

```
USE {Database-Initial-Name};
```

Then, start to create columns:

```
CREATE TABLE uploads (
  id INT AUTO_INCREMENT PRIMARY KEY,
  username VARCHAR(50) NOT NULL,
  filename VARCHAR(255) NOT NULL,
```

```
s3_url VARCHAR(512) NOT NULL,  
  
upload_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
  
);
```

To add an extra column, type:

```
ALTER TABLE uploads  
  
ADD COLUMN file_attachment VARCHAR(512) AFTER filename;
```

Then,

```
SHOW TABLES;
```

Run this:

```
DESCRIBE uploads;
```

You'll get this:

```
mysql> describe uploads;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default          | Extra          |  
+-----+-----+-----+-----+-----+-----+  
| id         | int           | NO   | PRI | NULL             | auto_increment |  
| username   | varchar(50)   | NO   |     | NULL             |                |  
| filename   | varchar(255)  | YES  |     | NULL             |                |  
| s3_url     | varchar(512)  | YES  |     | NULL             |                |  
| upload_date | timestamp     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |  
+-----+-----+-----+-----+-----+-----+  
5 rows in set (0.00 sec)  
  
mysql> 
```

NOTE:

- The id, filename, file_content, s3_url... are columns.
- Field, Type, Null, Key, etc., are headers for the columns. So if you want to modify columns, choose id, filename, etc.

To remove a column:

```
ALTER TABLE {table-name} DROP COLUMN {column-name};
```

To add a new column after an existing column:

```
ALTER TABLE {table-name}
```

```
ADD COLUMN {column-name} VARCHAR(20) AFTER {previous-column-name};
```

To show the contents of each column:

```
SELECT * FROM {table-name};
```

To show a specific column:

```
SELECT filename,file_content FROM {table-name};
```

To delete a specific row from a column:

```
DELETE FROM {table-name} WHERE id = {row-number};
```

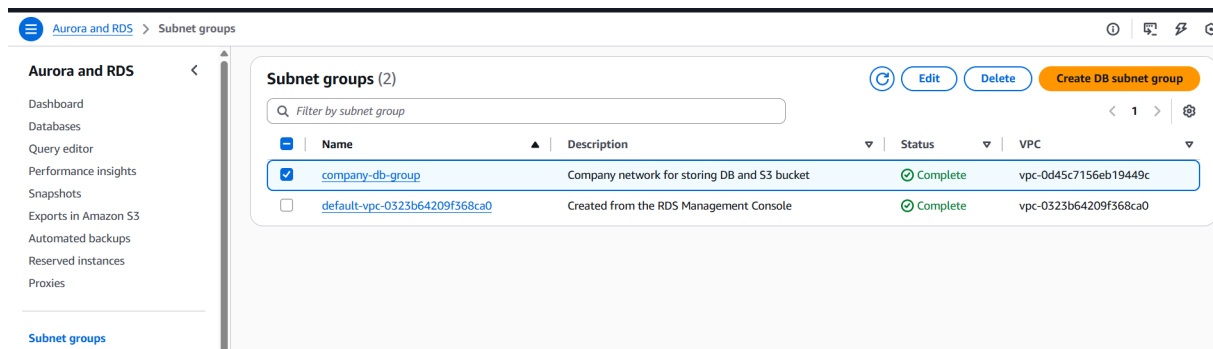
```
mysql> select * from uploads;
+----+-----+-----+-----+
| id | filename | s3_url          | upload_date      |
+----+-----+-----+-----+
| 1  | AdminUser | Screenshot (416).png | 0000-00-00 00:00:00 |
+----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> DELETE FROM uploads WHERE id =1 ;
Query OK, 1 row affected (0.01 sec)

mysql> select * from uploads;
Empty set (0.01 sec)

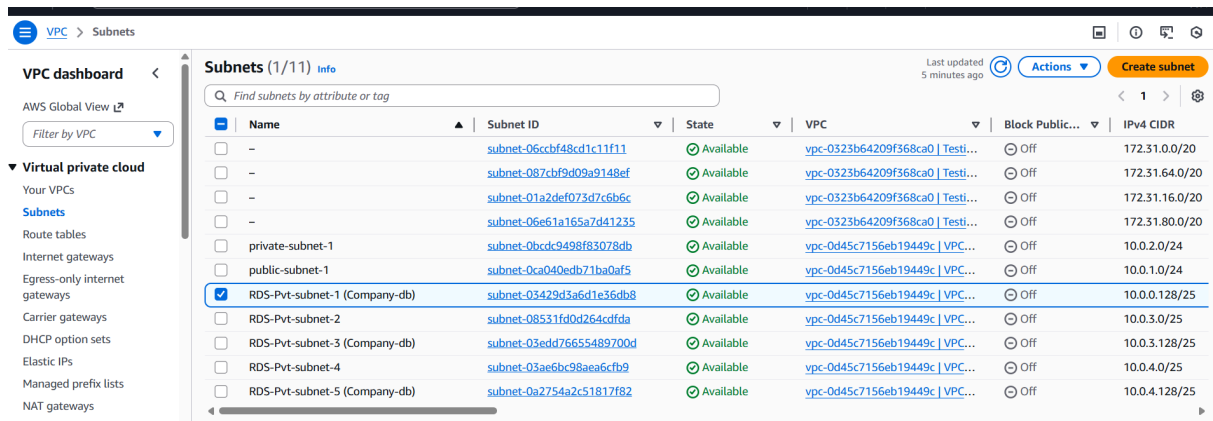
mysql> 
```

You may be confused about the DB Subnet Group in the RDS console.



Is it different from other subnets? Is it really a private subnet? How is it private? What if it's not?

> You can check in VPC/Subnet, you will see this.



Meaning, no. It's the same thing. Just different names from different directories. The DB subnet group still falls under the Subnet from the VPC console.

The databases should not talk directly to S3 buckets. We use BLOB (Binary Large Object).

Note: this is only for testing and will be deleted when we get into the communication.


```
mysql> describe uploads;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default          | Extra          |
+-----+-----+-----+-----+-----+-----+
| id         | int           | NO   | PRI | NULL             | auto_increment |
| username   | varchar(50)   | NO   |     | NULL             |                |
| filename   | varchar(255)  | YES  |     | NULL             |                |
| s3_url     | varchar(512)  | YES  |     | NULL             |                |
| upload_date | timestamp     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)

mysql> ALTER TABLE uploads
-> ADD COLUMN file_content LONGBLOB AFTER filename;
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> describe uploads;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default          | Extra          |
+-----+-----+-----+-----+-----+-----+
| id         | int           | NO   | PRI | NULL             | auto_increment |
| username   | varchar(50)   | NO   |     | NULL             |                |
| filename   | varchar(255)  | YES  |     | NULL             |                |
| file_content | longblob      | YES  |     | NULL             |                |
| s3_url     | varchar(512)  | YES  |     | NULL             |                |
| upload_date | timestamp     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> 
```

Private Instance Communication with S3 & DB

We need the Web Server / private instance to talk to the Database and S3. S3 and the database should not talk to each other.

- S3 is for storage.
- The database is for metadata.

Why not S3 talk directly to the database?

- The database will be slow cuz it will handle user uploads and call the API with the S3 bucket. It's like 2 works at once.
- SOC will have a hard time knowing what happens. It's better to know WHO uploads from the Web server, rather than the Database uploads something into the S3.
- If the Database is compromised, the S3 will likely to be compromised as well.
- A better question, where do the AWS credentials live? Database? Nah. It will be stolen and both the database and S3 will be compromised.

What about security? We will use an [IAM](#) Role feature and set it up for both the database and S3.

The web server will act as a middleman.

Creating S3

For creating S3, it's easy. Just watch YouTube.

IAM Roles

IAM Role Console: [Link](#)

The screenshot shows the 'Create role' page in the AWS IAM console. The left sidebar indicates the current step is 'Add permissions'. The main content area is divided into two sections: 'Trusted entity type' and 'Use case'.

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

Choose a use case for the specified service.

Use case

- ☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.
- ☐ **EC2 Role for AWS Systems Manager**

Permissions policies (1/1111) [Info](#)

Choose one or more policies to attach to your new role.

Filter by Type: All types (1 match)

Policy name	Type	Description
☒ AmazonS3FullAccess	AWS managed	Provides full access to all buckets via the ...

► Set permissions boundary - optional

Cancel Previous Next

☰ IAM > Roles > Create role

Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Private-Web-Server-To-S3-Role

Maximum 64 characters. Use alphanumeric and '+=, @-_' characters.

Description
Add a short explanation for this role.

Allows my private web instance to talk to S3 via the VPC Endpoint

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-_\[\]!#\$%^&*(){};:'"`,~. /

Review and click Create Role.

🔧 Step 2: Attach the Role to your EC2 Instance

1. Go to the **EC2 Console** and click on **Instances (running)**.
2. Select your **Private Web Instance** (the one that will host your website).
3. Click **Actions** (top right) > **Security** > **Modify IAM role**.
4. In the dropdown, select the role you just created: **Private-Web-Server-S3-Role**.
5. Click **Update IAM role**.

☰ EC2 > Instances

Instances (1/3) [Info](#)

Find Instance by attribute or tag (case-sensitive) Running

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
n8n-Instance	i-0d010ef71b5ac0cdb	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1f...
Load-Balancer	i-03035f16a3cd073d6	Running	t3.micro	3/3 checks passed	View alarms +	ec2-3...
☒ Private-Server	i-0cb6dfdc31bfb74e0	Running	t3.micro	3/3 checks passed	View alarms +	

i-0cb6dfdc31bfb74e0 (Private-Server)

☰ EC2 > Instances > i-0cb6dfdc31bfb74e0 > Modify IAM role

Modify IAM role [Info](#)

Attach an IAM role to your instance.

Instance ID
i-0cb6dfdc31bfb74e0 (Private-Server)

IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

Private-Web-Server-To-S3-Role Create new IAM role

Cancel Update IAM role

- Then click Update IAM Role.
- Go back to Private Server CLI via Public Instance machine.
- Type: `aws --version`
- We need to use AWS CLI for this because we are using an AWS service (IAM role) to help.
- If not yet installed:

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
```

```
sudo apt install unzip -y
```

```
unzip awscliv2.zip
```

```
sudo ./aws/install
```

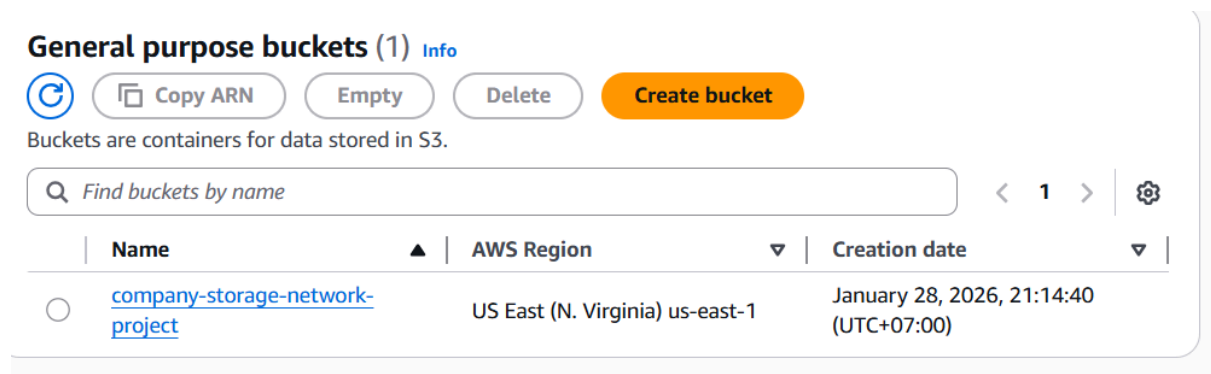
- **Result:**

```
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/WHEEL
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
  creating: aws/dist/wheel-0.46.3.dist-info/licenses/
inflating: aws/dist/wheel-0.46.3.dist-info/REQUESTED
inflating: aws/dist/wheel-0.46.3.dist-info/direct_url.json
inflating: aws/dist/wheel-0.46.3.dist-info/METADATA
inflating: aws/dist/wheel-0.46.3.dist-info/WHEEL
inflating: aws/dist/wheel-0.46.3.dist-info/entry_points.txt
inflating: aws/dist/wheel-0.46.3.dist-info/RECORD
inflating: aws/dist/wheel-0.46.3.dist-info/INSTALLER
inflating: aws/dist/wheel-0.46.3.dist-info/licenses/LICENSE.txt
You can now run: /usr/local/bin/aws --version
ubuntu@ip-10-0-2-37:~$ aws --version
aws-cli/2.33.13 Python/3.13.11 Linux/6.14.0-1018-aws exe/x86_64.ubuntu.24
ubuntu@ip-10-0-2-37:~$
```

- Check S3 bucket: `aws s3 ls`

```
ubuntu@ip-10-0-2-37:~$ aws s3 ls
2026-01-28 14:14:40 company-storage-network-project
ubuntu@ip-10-0-2-37:~$
```

Confirmation from S3 bucket console:



What is this IAM Role exactly?

Think of your Private Instance as a **person** and the IAM Role as a **Security Badge**.

- **Before the Role:** Your instance was just a "guest." When it tried to talk to S3, S3 asked, "Who are you?" Since the instance had no credentials, S3 slammed the door.
- **With the Role:** You didn't give the instance a username or password. Instead, you told AWS: *"This specific EC2 instance is allowed to wear the 'Private-Web-Server-S3-Role' badge."*
- **How it works (The Magic):** When you run `aws s3 ls`, the CLI looks at a hidden local "Metadata Service" (\$169.254.169.254\$). It pulls a **temporary** security token. S3 sees this token, recognizes it belongs to your role, and lets you in.

Why is this better? If a hacker steals your code, there are no S3 passwords in it for them to find. The "key" only exists inside the server itself.

S3 Interaction

Now that the "handshake" is working, you can do almost anything from the command line. Here are the most common "Company Network" commands to try:

A. Upload a test file:

Create a dummy file and push it to the bucket.

```
echo "This is a secret company file" > test.txt
```

```
aws s3 cp test.txt s3://YOUR-BUCKET-NAME/
```

B. List files in the bucket:

```
aws s3 ls s3://YOUR-BUCKET-NAME/
```

C. Download the file back:

```
aws s3 cp s3://YOUR-BUCKET-NAME/test.txt downloaded_test.txt
```

D. Delete the file:

```
aws s3 rm s3://YOUR-BUCKET-NAME/test.txt
```

How does this fit the Website Project

In a few minutes, when we write your Python/Flask website, the code will look like this:

1. **User** uploads a file to your **Web UI**.
 2. **Web UI** (using the IAM Role) sends that file to **S3**.
 3. **Web UI** then sends a message to **MySQL** saying: *"Hey, John just uploaded 'test.txt'. You can find it at this S3 Link."*
-



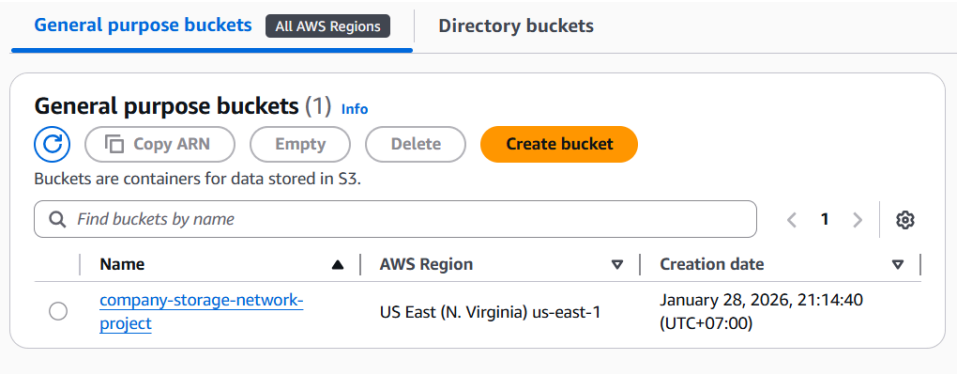
The Architecture Summary

You now have a "**Trusted Environment**":

- **MySQL** trusts the instance because of the **Security Group**.
- **S3** trusts the instance because of the **IAM Role**.
- **The Internet** can't see anything because the instance is **Private**.

Check if S3 is actually private

- Go to your bucket



- Click testRole.txt

☐	Screenshot 2026-01-26 202913.png	png	January 28, 2026, 21:21:02 (UTC+07:00)	54.3 KB	Standard
☐	testRole.txt	txt	February 3, 2026, 21:28:47 (UTC+07:00)	169.0 B	Standard

- Click the URL in the lower right corner

testRole.txt Info

PropertiesPermissionsVersions

Object overview

Owner
922e0812d260a66e60a60155ea23f65ba9bcb52861e2eae04d4f1710ce80d7

AWS Region
US East (N. Virginia) us-east-1

Last modified
February 3, 2026, 21:28:47 (UTC+07:00)

Size
169.0 B

Type
txt

Key
 testRole.txt

S3 URI
 s3://company-storage-network-project/testRole.txt

Amazon Resource Name (ARN)
 arn:aws:s3:::company-storage-network-project/testRole.txt

Entity tag (Etag)
 2fbc7741d86592e49a3052152d3135c4

Object URL
 <https://company-storage-network-project.s3.us-east-1.amazonaws.com/testRole.txt>

- Result in “Access Denied”
 - This means only the private instance is allowed to communicate. Currently, I am using a different IP and no role.

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes" ?>
<Error>
  <Code>AccessDenied</Code>
  <Message>Access Denied</Message>
  <RequestId>3NDRRHJ72FNAY696</RequestId>
  <HostId>Q9HCxQIkBSPwnp2R0fDeL/CuCnTDMmqJBi4+1C3q3FGz81/34wy+vBMWhDjrQ+gks0cCCBb0Uwss8yhyamUBozi02xhPlx79</HostId>
</Error>
```

Example:

```
ubuntu@ip-10-0-2-37:~$ aws s3 ls
2026-01-28 14:14:40 company-storage-network-project
ubuntu@ip-10-0-2-37:~$ aws s3 cp testRole.txt s3://company-storage-network-project
upload: ./testRole.txt to s3://company-storage-network-project/testRole.txt
ubuntu@ip-10-0-2-37:~$ aws s3 ls
2026-01-28 14:14:40 company-storage-network-project
ubuntu@ip-10-0-2-37:~$ aws s3 ls s3://company-storage-network-project/
2026-01-28 14:21:02      55625 Screenshot 2026-01-26 202913.png
2026-02-03 14:21:22      169 testRole.txt
ubuntu@ip-10-0-2-37:~$ aws s3 ls s3://company-storage-network-project/%
ubuntu@ip-10-0-2-37:~$ aws s3 cp s3://company-storage-network-project/testRole.txt testRole.txt
download: s3://company-storage-network-project/testRole.txt to ./testRole.txt
ubuntu@ip-10-0-2-37:~$ ls
aws  awscli2.zip  testRole.txt
ubuntu@ip-10-0-2-37:~$ aws s3 rm s3://company-storage-network-project/testRole.txt
delete: s3://company-storage-network-project/testRole.txt
ubuntu@ip-10-0-2-37:~$ aws s3 ls s3://company-storage-network-project/
2026-01-28 14:21:02      55625 Screenshot 2026-01-26 202913.png
ubuntu@ip-10-0-2-37:~$
```

Questions

Have you ever wondered why there are 2 security groups?

This is from the security group console.

☒	Company-DB-to-EC2 (using)	sg-01d146b4868dffc4d	rds-ec2-2	vpc-0d45c7156eb19449c
☒	Company-EC2-to-DB (using)	sg-0c09f26904c2d6f9b	ec2-rds-2	vpc-0d45c7156eb19449c

What you're seeing is the aftermath of the **AWS "Connect to EC2" Wizard**. When you use that wizard to link a database and a server, AWS doesn't just create one security group; it creates a **pair** to act as a "handshake."

To answer your question: **Yes, there is a big difference in their purpose**, even if their names look similar.

The "rds-ec2" vs. "ec2-rds" Breakdown

The naming convention tells you which way the "arrow" of permission is pointing:

SG Name Prefix	Where it is attached	What it does
rds-ec2	Attached to your RDS Database	The Lock: It opens port 3306. It says, "I allow traffic <i>from</i> anyone wearing the ec2-rds badge."

		It also has inbound rules. Meaning allowing from outside to come in.
ec2-rds	Attached to your EC2 Instance	The Badge: It doesn't actually have inbound rules. It has an outbound rule. Meaning connecting or communicating with the outside. It exists so the Database can "see" that this EC2 is an authorized friend.

This is from the Private Instance that wears the ec2-rds badge.

i-0cb6dfdc31bfb74e0 (Private-Server)

Details | Status and alarms | Monitoring | **Security** | Networking | Storage | Tags

▼ Security details

IAM Role
 Private-Web-Server-To-S3-Role [i](#)

Security groups
 sg-0c09f26904c2d6f9b (ec2-rds-2)
 sg-0e6ee758106ebfc38 (Private-Server)

Owner ID
 967883358429

Launch time
Mon Jan 26 7

This is from the Company-DB that wears the rds-ec2 badge.

Connectivity & security

Endpoint & port

Endpoint
 company-db.c2lqwc4e8qkg.us-east-1.rds.amazonaws.com

Port
3306

Networking

Availability Zone
us-east-1a

VPC
[VPC-Company-Production-Network \(vpc-0d45c7156eb19449c\)](#)

Subnet group
company-db-group

Subnets
[subnet-0a2754a2c51817f82](#)
[subnet-03429d3a6d1e36db8](#)
[subnet-03edd76655489700d](#)

Network type
IPv4

Security

VPC security groups
[rds-ec2-2 \(sg-01d146b4868dffa4\)](#)
 Active

Publicly accessible
No

Certificate authority [Info](#)
rds-ca-rsa2048-g1

Certificate authority date
May 26, 2061, 06:34 (UTC+07:00)

DB instance certificate expiration date
January 29, 2027, 22:40 (UTC+07:00)

=====

ENDS